



Douglas Electrical Components

HERMETIC MIL-DTL-38999 CONNECTORS



MOISTURE-TIGHT, HERMETICALLY SEALED CONNECTORS IN 4-6 WEEKS

FEATURES

- Series I & III Circular Connectors
- Choose from standard box mount or jam nut mounting options
- Integrate solder cups, PC tails, or wire leads

APPLICATIONS

- Military & Aerospace
- Instrumentation & Controls
- Oil & Gas
- Industrial & IoT

OVERVIEW

Douglas Electrical Components' hermetic MIL-DTL-38999 connectors are designed for power, signal, and networking applications across various industries. Its compact and lightweight design, coupled with robust metal housing and various pin counts, offers design flexibility without compromising durability in harsh environments.

SPECIFICATIONS

Housing Material	Standard options include aluminum and stainless steel. See series level matrix for more details. Contact factory for other materials.
Sealing Material	ASTM E-595-93 Rated Hermetic Epoxy
Performance	
Helium Leak	< 1x10 ⁻⁸ cc He/sec. Contact factory for more options.
Pressure Rating	Up to 1 atm standard; higher pressures available upon request
Operating Temperature Range	-40°F to 225°F (-40°C to 107°C) Consult factory for expanded thermal range assemblies up to 200°C
Certifications	
Quality System	ISO 9001:2008; ISO140001
Workmanship	IPC-620 certified operators
ITAR	ITAR-DDTC; contact factory for ITAR regulated designs.
Traceability	NIST traceability; vendor traceability available upon request
Design and Manufacturing Site	Randolph, NJ USA

TEMPERATURE

Formula

°F

= [°R] -459.69

= [°C] 9/5+32

= ([°K] - 273.16) 9/5 + 32

°C

= [°K] -273.16

= ([°F] - 32)5/9

= ([°R] - 491.69) 5/9

°R

= [°F] -459.96

= [°C] -9/5 +491.69

= ([°K] - 273.16) 9/5+ 491.69

°K

= [°C] - 273.16

= ([°F] - 32) 5/9+273.16

= ([°R] - 491.69) 5/9+ 273.16

Quick Conversion

°C	°F
-50	-58
-40	-40
0	32
25	77
105	221
125	257
135	275
150	302
175	347
200	392

Cryogen Boiling Points

	°F	°C	°R	°K
He	-452.1	-268.9	7.6	4.22
H ₂	-423.2	-252.8	36.5	20.3
N ₂	-320.5	-195.8	139.2	77.1
Air	-380.9	-194.4	141.8	78.8
O ₂	-297.2	-182.9	162.3	90.3

Absolute Zero

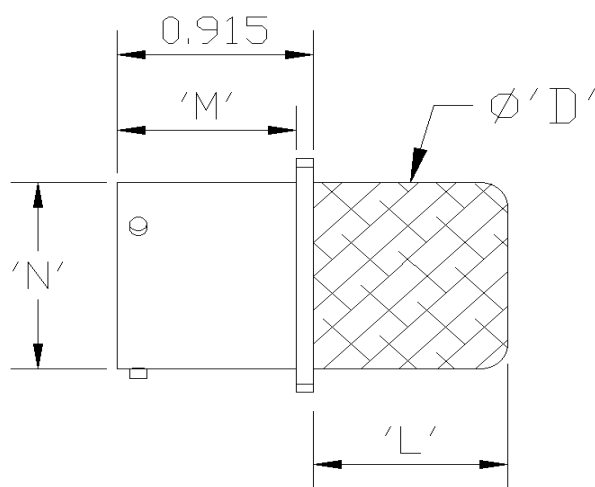
-459.69°F
-273.16°C
0°R
0°K

INSERT ARRANGEMENTS

D38999 Series I Insert Arrangement	22D	20	16	12
09-35	6			
09-98		3		
11-02			2	
11-05		5		
11-35	13			
11-98		6		
11-99		7		
13-04			4	
13-08		8		
13-35	22			
13-98		10		
15-05			5	
15-18		18		
15-19		19		
15-35	37			
17-06				6
17-08			8	
17-26		26		
17-35	55			
19-11			11	
19-32		32		
19-35	66			
21-11				11
21-16			16	
21-35	79			
21-41		41		
23-21			21	
23-35	100			
23-53		53		
23-55		55		
25-19				19
25-29			29	
25-35	128			
25-37			37	
25-61		61		

D38999 Series III Insert Arrangement	22D	20	16	12
A35	6			
A98		3		
B2			2	
B4		4		
B5		5		
B35	13			
B98		6		
B99		7		
C4			4	
C8		8		
C35	22			
C98		10		
D5			5	
D18		18		
D19		19		
D35	37			
E6				6
E8			8	
E26		26		
E35	55			
F11			11	
F32		32		
F35	66			
F45	67			
G11				11
G16			16	
G24		24		
G25		25		
G27		27		
G35	79			
G41		41		
H21			21	
H32		32		
H34		34		
H35	100			
H36		36		
H53		53		
H55		55		
H97			16	
H99			11	
J19				19
J29			29	
J35	128			
J61		61		

38999 SERIES I BOX MOUNT



DIMENSIONAL DATA

SHELL SIZE	N	M	L	D
9	0.5702	0.820	1.07	0.60
11	0.700	0.820	1.07	0.72
13	0.850	0.820	1.07	0.85
15	0.975	0.820	1.07	0.97
17	1.100	0.820	1.07	1.10
19	1.207	0.820	1.07	1.22
21	1.332	0.790	1.07	1.35
23	1.457	0.790	1.07	1.47
25	1.582	0.790	1.07	1.60

ORDERING INFORMATION

58657

Housing Material

B = Aluminum Shell, olive drab cadmium over nickel base

F = Aluminum Shell, Electroless nickel finish

A = Aluminum Shell, silver to light iridescent yellow color (bright) cadmium over electroless nickel

X - XXXX - XX - XXX

Electrical Connection

SC = Soldercup

PC = PC Tail

WR = Wire

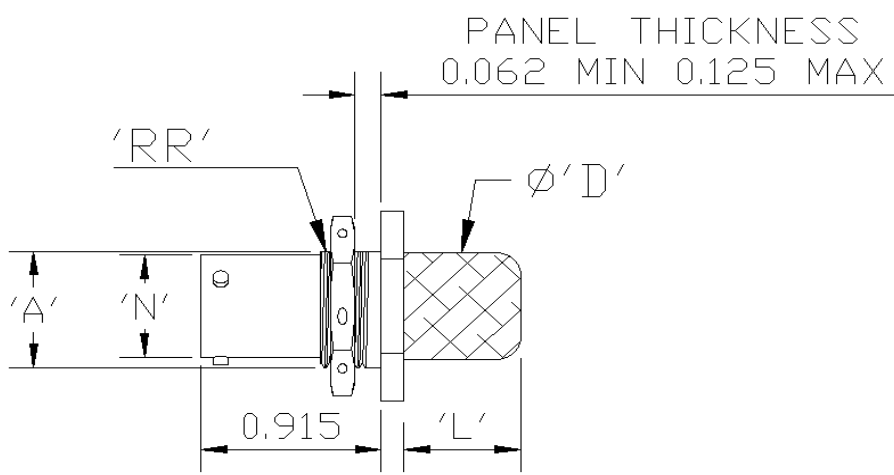
Length

(e.g. -30) = Length in inches

Insert Arrangement

Insert Arrangement (e.g. 1335 for a 13-35 insert)

38999 SERIES I JAM NUT



DIMENSIONAL DATA

SHELL SIZE	N	A	RR THREAD	L	D
9	0.570	0.669	.6875-24 UNEF	0.754	0.60
11	0.698	0.769	.8125-20 UNEF	0.754	0.72
13	0.848	0.955	1.0000-20 UNEF	0.754	0.85
15	0.973	1.084	1.1250-18 UNEF	0.754	0.97
17	1.098	1.208	1.2500-18 UNEF	0.754	1.10
19	1.205	1.333	1.3750-18 UNEF	0.723	1.22
21	1.330	1.459	1.5000-18 UNEF	0.723	1.35
23	1.455	1.580	1.6250-18 UNEF	0.723	1.47
25	1.580	1.709	1.75000-18 UNEF	0.723	1.60

ORDERING INFORMATION

58658

X - XXXX - XX - XXX

Housing Material

- B = Aluminum Shell, olive drab cadmium over nickel base
- F = Aluminum Shell, Electroless nickel finish
- A = Aluminum Shell, silver to light iridescent yellow color (bright) cadmium over electroless nickel

Fourth “-XXX”

(e.g. -30) = Length in inches

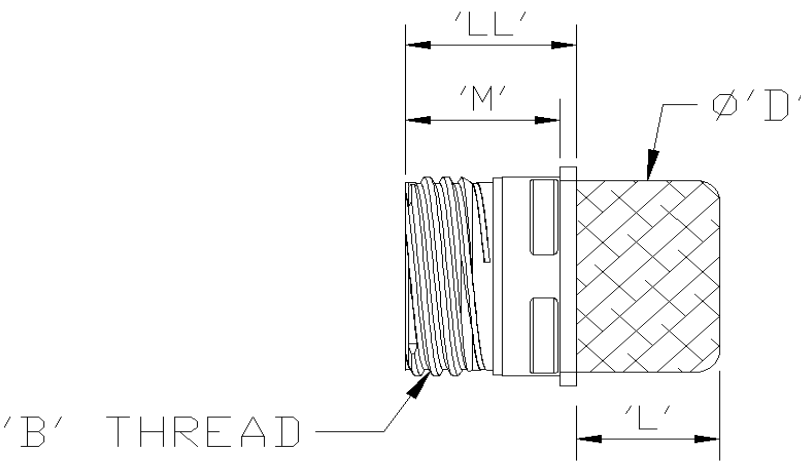
Third “-XX”

- SC = Soldercup
- PC = PC Tail
- WR = Wire

Second “-XXXX”

Insert Arrangement (e.g. 1335 for a 13-35 insert)

38999 SERIES III BOX MOUNT



DIMENSIONAL DATA

SHELL SIZE	LL	M	B THREAD CLASS 2A	L	D
9 (A)	0.905	0.820	0.625	0.92	0.63
11 (B)	0.905	0.820	0.750	0.92	0.75
13 (C)	0.905	0.820	0.875	0.92	0.87
15 (D)	0.905	0.820	1.000	0.92	1.03
17 (E)	0.905	0.820	1.188	0.92	1.14
19 (F)	0.905	0.820	1.250	0.92	1.26
21 (G)	0.905	0.790	1.375	0.92	1.38
23 (H)	0.905	0.790	1.500	0.92	1.50
25 (J)	0.905	0.790	1.625	0.92	1.62

ORDERING INFORMATION

58659

X - XXX - XX - XXX

Housing Material

- F = Aluminum Shell, Electroless Nickel Finish
- K = Stainless Steel Shell, Passivated - Firewall
- S = Stainless Steel Shell, electrodeposited nickel
- W = Aluminum Shell, olive drab cadmium over nickel base
- J = Composite Shell, olive drab cadmium
- M = Composite Shell, Electroless Nickel
- T = Nickel - PTFE (Aluminum) Durmalon
- G = Space Grade, Outgassed
- L = Corrosion-Resistant Steel/ Electrodeposited Nickel
- Z = Black Zinc Nickel

Fourth “-XXX”

(e.g. -30) = Length in inches

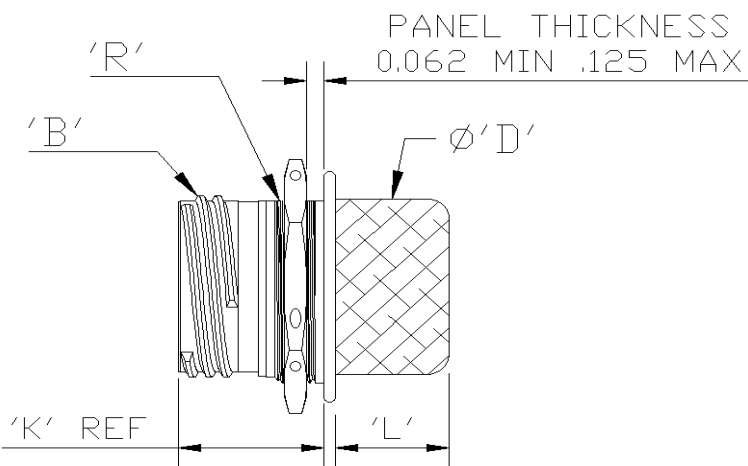
Third “-XX”

- SC = Soldercup
- PC = PC Tail
- WR = Wire

Second “-XXX”

Insert Arrangement
(e.g. C35 for a C35 insert)

38999 SERIES III JAM NUT



DIMENSIONAL DATA

SHELL SIZE	K	B THREAD CLASS 2A	R THREAD	L	D
9 (A)	0.871	0.625	M17x1-6g	0.82	0.63
11 (B)	0.871	0.750	M20x1-6g	0.82	0.75
13 (C)	0.878	0.875	M25x1-6g	0.82	0.87
15 (D)	0.878	1.000	M28x1-6g	0.82	1.03
17 (E)	0.878	1.188	M32x1-6g	0.82	1.14
19 (F)	0.878	1.250	M35x1-6g	0.82	1.26
21 (G)	0.878	1.375	M38x1-6g	0.82	1.38
23 (H)	0.878	1.500	M41x1-6g	0.82	1.50
25 (J)	0.878	1.625	M44x1-6g	0.82	1.62

ORDERING INFORMATION

58660

X - XXX - XX - XXX

Housing Material

- F = Aluminum Shell, Electroless Nickel Finish
- K = Stainless Steel Shell, Passivated - Firewall
- S = Stainless Steel Shell, electrodeposited nickel
- W = Aluminum Shell, olive drab cadmium over nickel base
- J = Composite Shell, olive drab cadmium
- M = Composite Shell, Electroless Nickel
- T = Nickel - PTFE (Aluminum) Durmalon
- G = Sapce Grade, Outgassed
- L = Corrosion-Resistant Steel/ Electrodeposited Nickel
- Z = Black Zinc Nickel

Fourth “-XXX”

(e.g. -30) = Length in inches

Third “-XX”

- SC = Soldercup
- PC = PC Tail
- WR = Wire

Second “-XXX”

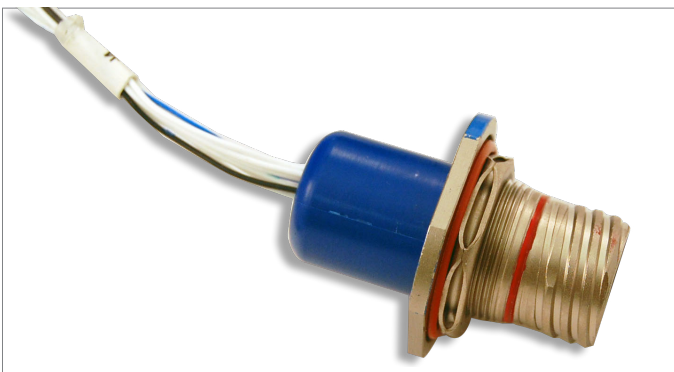
Insert Arrangement
(e.g. C35 for a C35 insert)

BEYOND THE CATALOG

Douglas Electrical Components specializes in the design and manufacture of customized hermetic connector solutions. Our hermetic technology permits the design flexibility to integrate virtually any conductor or optical fiber into a plug-and-play assembly. We can customize the electrical and mechanical interface to create a Design for Manufacturing (DFM) moisture-tight, hermetic solution for harsh environments and challenging applications.

DESIGN OPTIONS INCLUDE:

- Choose connection types from traditionally non-hermetic 38999 designs
- Integrate wires, cables, flex and rigid circuits, and optical fibers
- Customize the mechanical assembly in various materials, interfaces, and design features
- Add increased protection with shielding, sleeves, and markings



To Order

You may place an order directly with our sales staff at 973.627.8230.

Technical Questions or Design Assistance

Please call our Product Engineering Department 973.627.8230.

Standard Terms and Conditions

Domestic shipments: All products are shipped F.O.B. Randolph, NJ. We offer a prompt payment discount of 1% 10 days, Net Cash 30 days to customers who have had their credit approved. Prices and specifications are subject to change without notice. Foreign shipments: All terms and conditions will be negotiated at time of order placement.

Note to Purchaser

While we believe all statements, representations, and recommendations are correct and reliable, we cannot guarantee accuracy or completeness. It is the user's responsibility to determine and confirm suitability for use. All sales of Douglas Electrical Components' products are subject to our Standard

Terms and Conditions. A copy of our Standard Terms and Conditions may be obtained by calling our Sales Administration Office at 973.627.8230. Douglas Electrical will provide Special Certificates of Compliance, Test Reports, and certificates of origin but only if they are requested at the time of order.

Warranty

All products are warranted to be free of defects in materials and workmanship for a period of twelve months from date of shipment. Our liability, unless explicitly changed by us, is strictly limited to the repair or replacement (at our option) of defective parts. Any defective parts must be returned to us to receive credit or replacement. Returned product must be accompanied by an RMA number. RMA numbers may be obtained by calling our Sales Administration Office at 973.627.8230.

Errors

While we believe that all information in this catalog is accurate and correct, we do not take responsibility for typographical errors.



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